Software Requirements Specification

for

Kandypack Rail and Road-based Supply Chain Distribution System

Version 1.0

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for Version 1.0 of the **Kandypack Rail and Road-based Supply Chain Distribution System**. Kandypack is a mid-sized FMCG manufacturer based in Kandy, Sri Lanka, that distributes finished goods to wholesale and retail customers across the island using Sri Lanka Railways for bulk transport between Kandy and six destination cities (Colombo, Negombo, Galle, Matara, Jaffna, Trincomalee), followed by truck-based last-mile delivery from store facilities located near the destination railway stations.

This document covers the complete system: **order placement and validation, rail cargo scheduling, store receipt and inventory handling, truck/driver/assistant roster scheduling, last-mile delivery execution and tracking, business reporting, and user account management**. It is intended to replace Kandypack's existing Excel-based tracking process with a single database-driven web application.

1.2 Document Conventions

- Every functional requirement is labeled REQ-<feature>-<number> (e.g., REQ-ORD-01).
- **Bold text** highlights system modules, roles, or entities of particular importance.
- *Italic text* introduces a new or specialized term.
- Requirements inherit the priority of the System Feature section under which they appear, unless individually marked otherwise.
- "TBD" marks information not yet finalized by the team; all TBD items are collected in Appendix C.

1.3 Intended Audience and Reading Suggestions

This document is intended for the following readers:

- **Project Team / Developers** - to design and implement the system against these requirements.
- **Database Designers** - to derive the ER model, normalization, keys, triggers, procedures, and indexing strategy described in Sections 4 and 6.
- **Course Instructors / Evaluators** - to assess completeness and correctness of the requirements against the project brief.
- **Kandypack Stakeholders** (represented roles) - Admin, Dispatcher, Regional Store Manager, Driver, Assistant, and Customer - whose workflows are captured in Section 2.3 and Section 4.

Readers should begin with **Sections 1** and **2** for the project overview, then read **Section 3** for **interface expectations**, **Section 4** for the **detailed functional requirements** organized by system feature, and **Section 5** for **non-functional requirements** and **business rules**. **Section 6** for the Appendices contain supporting database, glossary, and open-items information.

1.4 Product Scope

The system is a centralized, database-driven logistics platform covering Kandypack's full distribution pipeline from order placement to proof of delivery.

1.4.1 Key Roles of the System

- **Islandwide Distribution Network:** Streamlines FMCG distribution originating from Kandy to destination hubs throughout Sri Lanka.
- **Bulk Rail Freight Coordination:** Manages primary long-haul transport by scheduling bulk cargo via the national railway network.
- **Last-Mile Delivery Logistics:** Directs fleet trucks to transport goods from rail terminals directly to final destinations.
- **Process Automation:** Replaces legacy manual workflows and spreadsheets with an integrated, centralized database management system.
- **Core Logistics Architecture:** Delivers end-to-end operational coverage across key functional areas:
    - Order Lifecycle Management
    - Rail Freight Scheduling
    - Fleet & Route Optimization
    - Crew & Driver Shift Assignment
    - Live Shipment Tracking & Status Updates
    - Operational Analytics & Executive Reporting
- **Proactive Risk Management:** Configures real-time notifications and system alerts to detect and resolve delivery disruptions early.
- **Regulatory & Workforce Compliance:** Automatically enforces national transport safety regulations and driver working-hour mandates.

1.4.2 Objectives and Benefits

- **Reduce logistical delays and operational errors.**
- **Improve visibility and traceability of shipments.**
- **Upgrade use of available transport resources.**
- **Improve customer satisfaction through timely deliveries.**

Out of scope for Version 1.0: **payment processing/invoicing, warehouse/inventory management beyond store-level receipt reconciliation, native mobile applications (a**

responsive web interface is used for all roles, including drivers and assistants), and international distribution.

1.5 References

- IEEE. IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications. IEEE Computer Society, 1998.
- "Software Requirements Specification Example - WebApp", Department of Computer Science and Engineering, Michigan State University.
- Project 3 - Rail and Road-based Supply Chain Distribution System, project brief document (course-provided).

2. Overall Description

2.1 Product Perspective

The Kandypack system is a new, self-contained, custom-built product. It replaces Kandypack's current Excel-based tracking with a single centralized web application used by all internal roles and by customers. The system does not integrate with an external live railway API in Version 1.0; instead, Kandy HQ administrators enter and maintain train trip schedules and capacities directly within the system, since Sri Lanka Railways does not currently expose a programmatic booking interface for cargo capacity.

The system follows a client-server architecture: a single backend/database serves role-specific views (dashboards) through one responsive web front end. The database is the single source of truth for orders, schedules, rosters, inventory movement, and reporting, which is what allows the ACID, trigger, and stored-procedure requirements in Section 6 to be enforced centrally rather than duplicated across role-specific tools.

2.2 Product Functions

- **Order Management**: order placement, 7-day lead-time validation, address-to-route matching, order history and status tracking (Placed -> Scheduled -> In Transit -> Delivered -> Closed).

- **Rail Cargo Scheduling**: allocation of order volume to train trips within fixed cargo capacity and per-product space rates, automatic overflow to the next trip, and train-loading confirmation.

- **Store Receipt and Truck Demand Planning**: confirmation of received cargo at each destination store, discrepancy flagging.

- **Truck/Driver/Assistant Rostering:** weekly assignment of trucks, drivers, and assistants to route runs, with automatic enforcement of nonoverlap, consecutive-run, and weekly-hour rules.

- **Delivery Execution and Tracking**: driver/assistant pickup and delivery status updates, and separate customer-side delivery confirmation.

- **Reporting and Analytics**: the six management reports listed in Section 1.4, each filterable and exportable.

- **User Management and Authentication**: role-based accounts for Admin, Dispatcher, Regional Store Manager, Driver, Assistant, and Customer.

- **Notifications and Alerts:** system-generated alerts for capacity delays, roster conflicts, and approaching working-hour limits.

2.3 User Classes and Characteristics

The system supports six user classes, aligned to Kandypack's actual operating structure rather than a single generic "staff" role. This distinction matters because each class has a different data scope (company-wide vs. one city vs. one truck run vs. one customer's own orders) and a different device/skill profile.

1. System Administrator (Kandy HQ / Management)

Company-wide, high-privilege role based at the Kandy main branch. Maintains master data: train trip schedules and capacities, the product catalog and each product's space-consumption rate, predefined delivery routes (coverage area and maximum delivery time) for each of the six cities, truck records, and driver/assistant employee records. Also manages user accounts across all roles and is the only role with unrestricted access to all six management reports. Computer-literate, moderate-to-high technical skill expected.

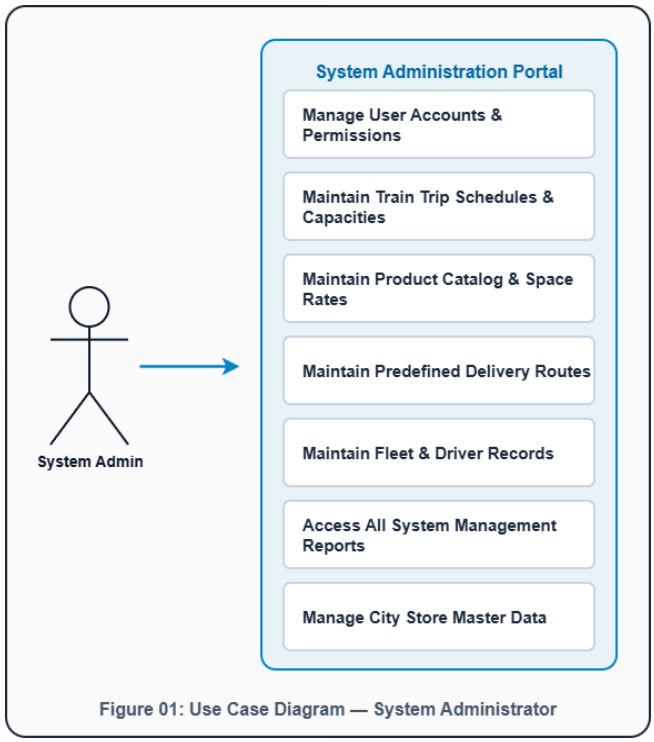


Figure 01: Use Case Diagram — System Administrator

2. Logistics Dispatcher (Kandy HQ)

Operates at Kandy HQ under the Administrator. Reviews orders awaiting bulk transport, allocates them to train trips within available cargo capacity, and marks a trip as loaded once goods are physically placed on the train. A week ahead of each delivery date, reviews the system's suggested truck/driver/assistant requirement per route (derived from historical order volume) and builds the weekly roster, resolving any conflicts the system flags. Moderate computer skill, working knowledge of logistics scheduling.

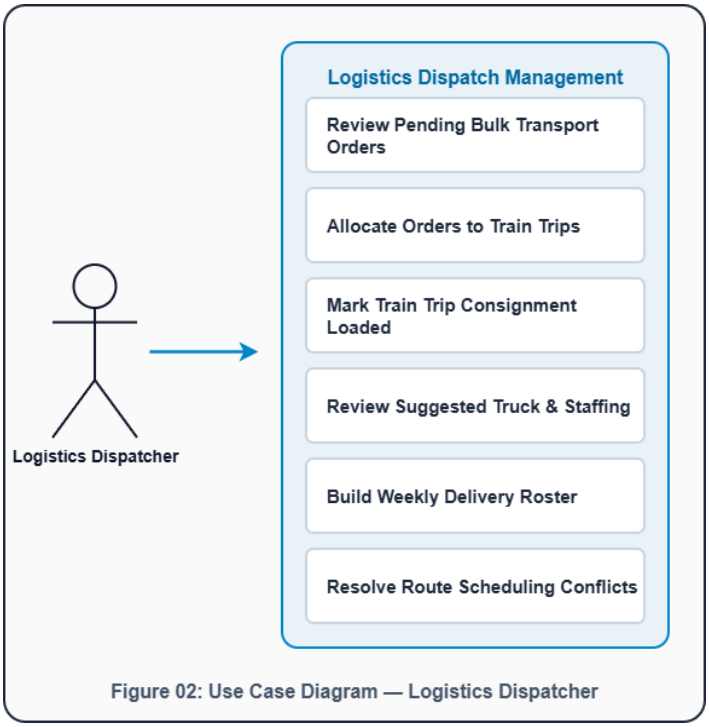


Figure 02: Use Case Diagram — Logistics Dispatcher

3. Regional Store Manager (one per destination city)

Based at the store facility adjacent to the railway station in Colombo, Negombo, Galle, Matara, Jaffna, or Trincomalee. Confirms arrival and receipt of each train trip's consignment for their city, reconciles received quantities against the manifest and flags discrepancies, and oversees loading of received orders onto that day's truck runs for the routes covering their city. Access is restricted to their own city's data. Computer-literate, moderate technical skill.

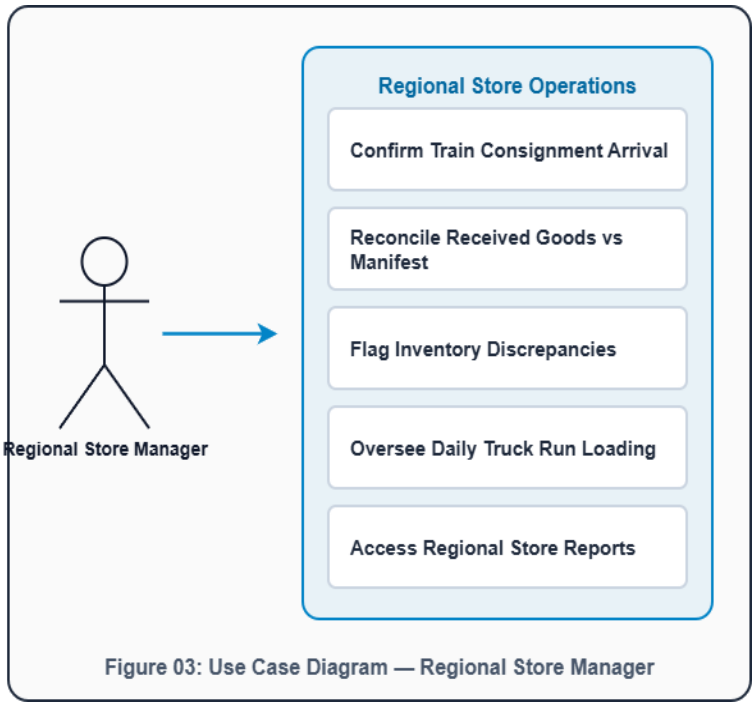


Figure 03: Use Case Diagram — Regional Store Manager

4. Truck Driver

Field role using a simplified, mobile-optimized subset of the same web application. Views the day's assigned run (route, truck, assistant, and stop list), confirms pickup from the store, and confirms or reports the outcome of each delivery stop. Minimal technical skill assumed; interface favors large touch targets and few required fields over dense tables.

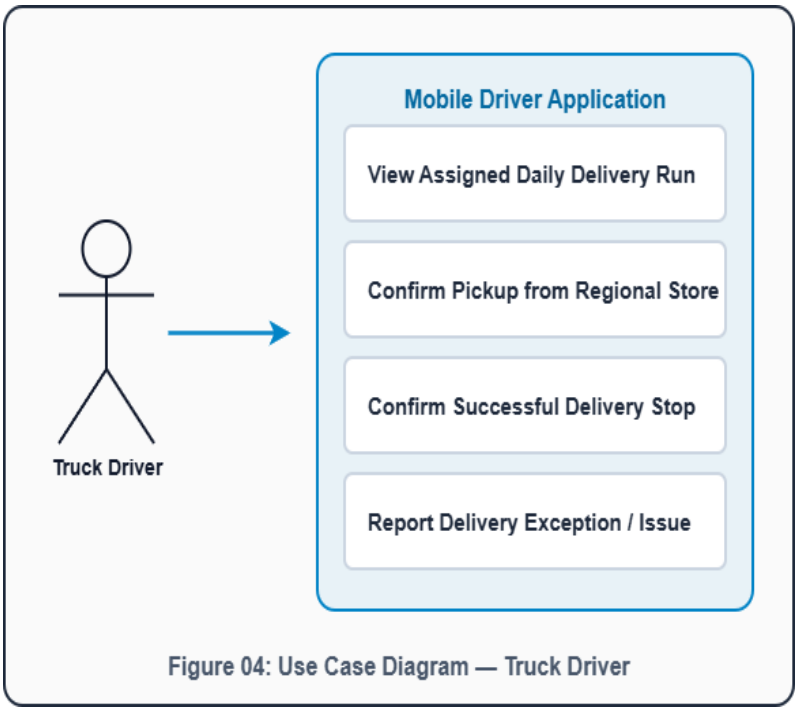


Figure 04: Use Case Diagram — Truck Driver

5. Delivery Assistant

Paired with a driver on a truck run, using the same simplified mobile interface. May update pickup/delivery status alongside the driver. Assistants are subject to a looser consecutive-assignment limit (up to two consecutive routes) and a higher weekly hour cap (60 hours) than drivers, reflecting the different labor rule in the project brief.

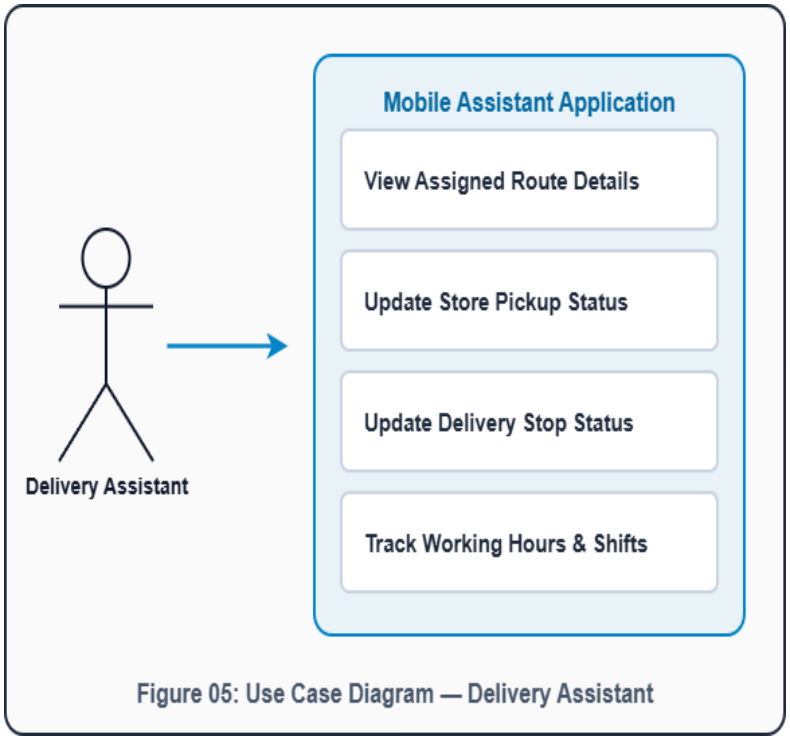


Figure 05: Use Case Diagram — Delivery Assistant

5. Customer

External user (wholesale or retail buyer) who self-registers, places multi-item orders against a delivery address, views their own order history and live status, and separately confirms receipt once a driver has marked a delivery complete. Can modify or cancel an order only before it has been assigned to a train trip. Minimal technical skill assumed; may access via desktop or mobile browser.

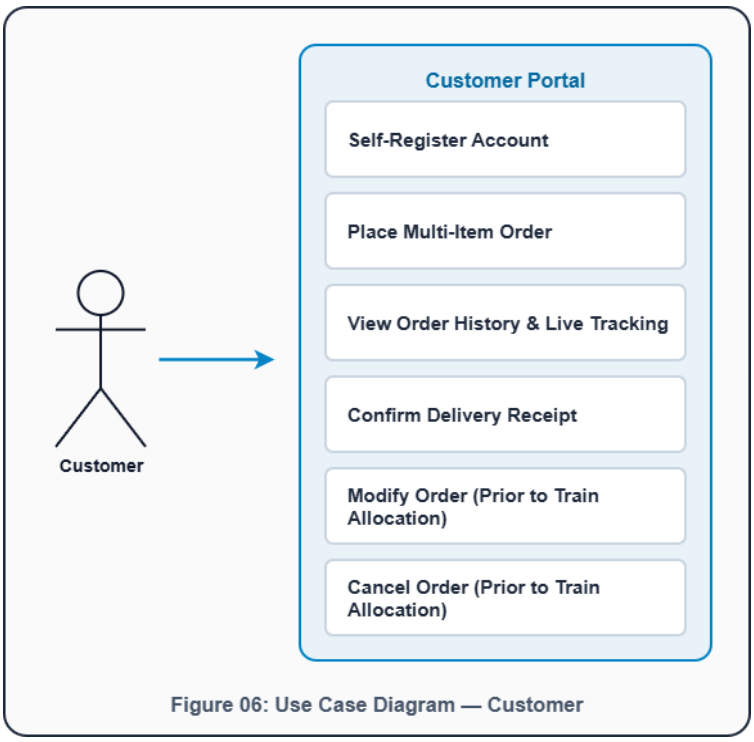


Figure 06: Use Case Diagram — Customer

2.4 Operating Environment

The system operates as a client-server web application. The database server uses MySQL. Clients (Admin, Dispatcher, Regional Store Manager, Driver, Assistant, Customer) access the system through a standard web browser on desktop, tablet, or mobile devices; no native mobile application is required for Version 1.0, since the Driver/Assistant views are a responsive subset of the same application. Field use by drivers/assistants assumes intermittent mobile data connectivity.

2.5 Design and Implementation Constraints

- The system must be implemented using open-source technologies to minimize licensing cost.
- The relational schema must support ACID transactions, foreign-key constraints, indexing, and stored procedures/triggers, per the project brief and Section 6.
- Sri Lanka Railways does not expose a live capacity API for Version 1.0; train schedules and capacities are administrator-entered and must be treated as authoritative within the system.
- The system must comply with Sri Lankan data protection expectations for customer personal data (name, address, contact details).
- The schema must allow modular extension (e.g., warehouse/inventory management, payment processing) in future versions without a full redesign.

2.6 User Documentation

- A role-based User Guide covering the Admin, Dispatcher, Regional Store Manager, and Customer workflows.
- A one-page Quick Start Guide for the Driver/Assistant mobile view.
- In-application help text/tooltips on the order placement and scheduling screens.

2.7 Assumptions and Dependencies

- Product physical dimensions and space-consumption rates are assumed to be provided by Kandypack's product team; where unavailable, the team will take reasonable illustrative assumptions (e.g., 1 standard carton of detergent consumes 0.5 space units) for population/testing data.
- Train trip frequency and cargo capacity per trip are assumed to be fixed values entered by the Administrator, not sourced from a live railway API.
- Delivery routes are assumed to be pre-drawn, non-overlapping coverage areas per city, each with an agreed maximum delivery time; route geometry itself is out of scope (the system stores route metadata and a served-address/area rule, not a mapping engine).
- Users have access to reasonably reliable internet at Kandy HQ, the regional stores, and via mobile data in delivery vehicles.
- Kandypack's transport fleet, dispatch staff, and regional store staff will be available for requirements validation and pilot testing.
- The project brief does not specify a break duration for the rule that a driver must not be scheduled for two consecutive truck deliveries. The team has adopted a minimum 2-hour rest break between any two runs assigned to the same driver on the same day as a reasonable working assumption, consistent with typical commercial-transport rest practices; this value can be revised without changing the system design, since it is stored as a configurable rule rather than hard-coded.

3. External Interface Requirements

3.1 User Interfaces

UI mockups represent the intended design direction, noting that final visual design, colors, logos, and branding are not final and may change during implementation.

All interfaces are delivered through a single web application; each role sees a dashboard scoped to its permissions and data. Administrator, Dispatcher, and Regional Store Manager use a desktop-oriented sidebar layout that remains responsive down to tablet width. Driver, Assistant, and Customer use a lightweight, mobile-first layout instead; figures for those roles are explicitly labeled "(Mobile view)" and were captured at a phone-width viewport to show the layout as it appears on a handset.

3.1.1 Login and Registration

- Login: email/username and password, with role-based redirect to the appropriate dashboard on success; "Invalid username or password" style errors shown inline.
- Customer Registration: name, email (used as username), password with strength validation, confirm password, and delivery address; internal accounts (Admin, Dispatcher, Regional Store Manager, Driver, Assistant) are created by the Administrator rather than self-registered.

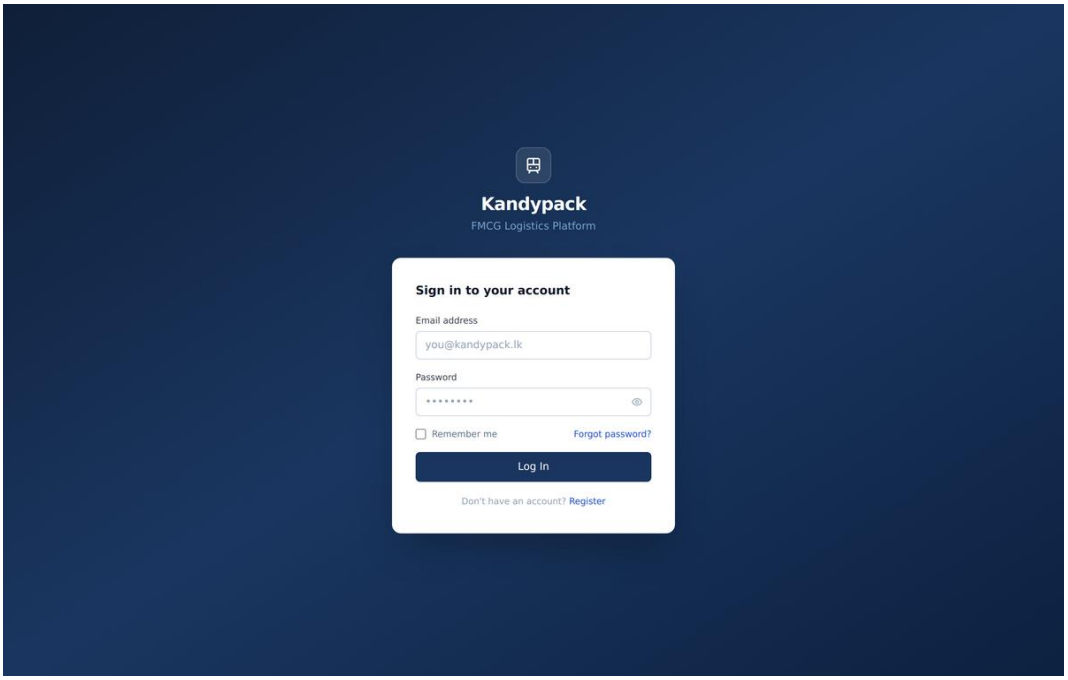


Figure 1: Login screen, with role-based quick-access

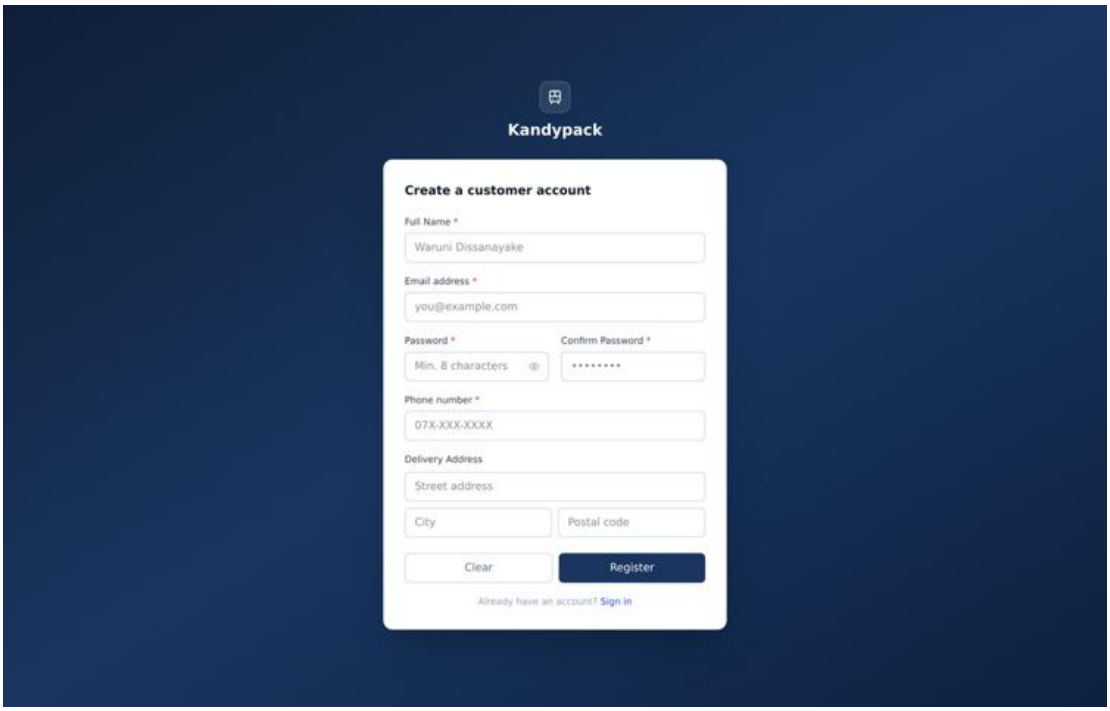


Figure 2: Customer registration screen

3.1.2 Administrator Interfaces

- Master Data screens for train schedules, product catalog and space rates, delivery routes, trucks, and employee (driver/assistant) records, each with add/edit/deactivate actions.
- User Management screen: list, add, edit role/permissions, and deactivate user accounts.
- Report Overview screen giving access to all six report types with date-range, city, route filters and CSV/PDF export.

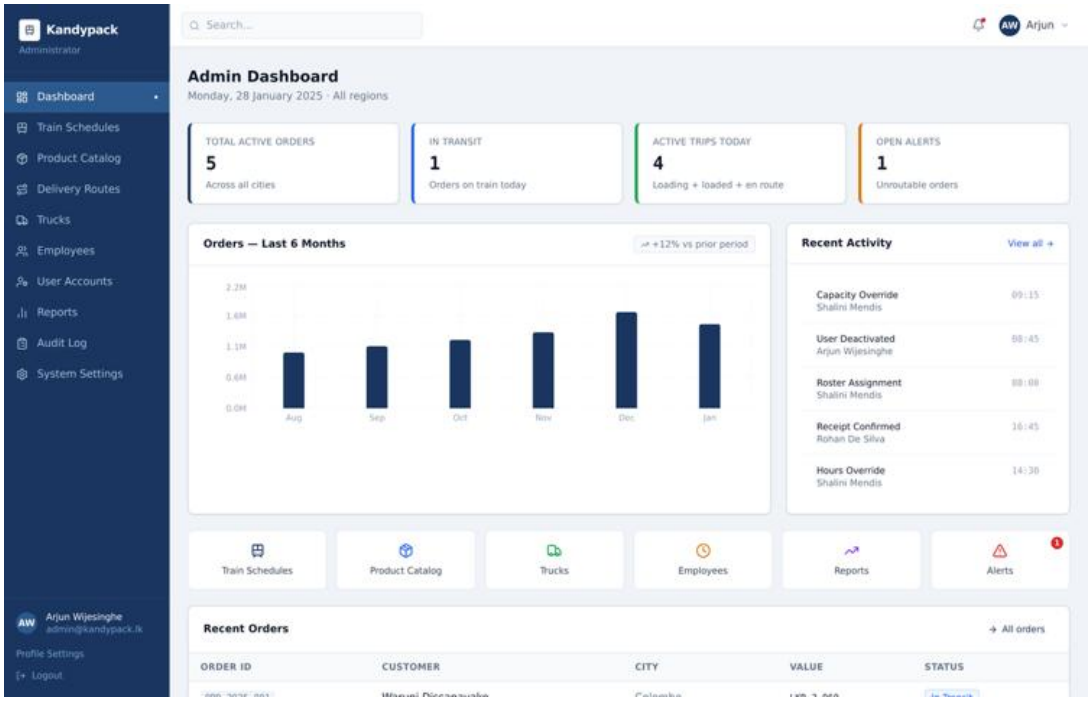


Figure 3: Administrator Dashboard (home screen), showing company-wide summary cards, order trends, and recent activity

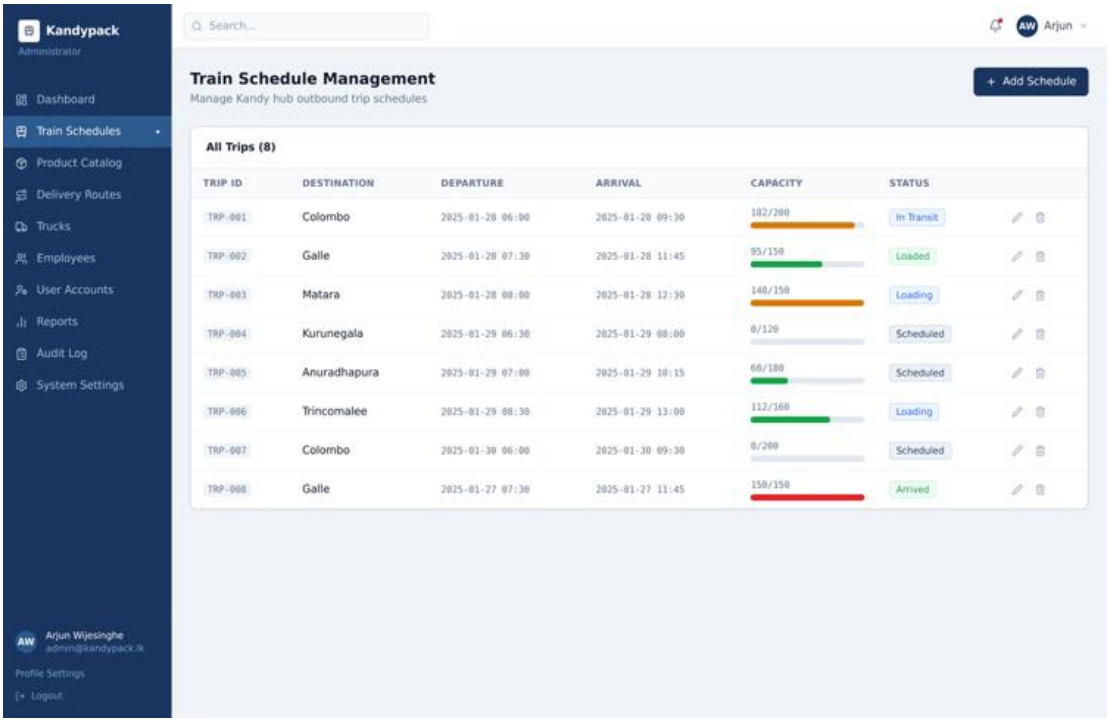


Figure 4: Train Schedule Management screen, representative of the Administrator's master-data screens

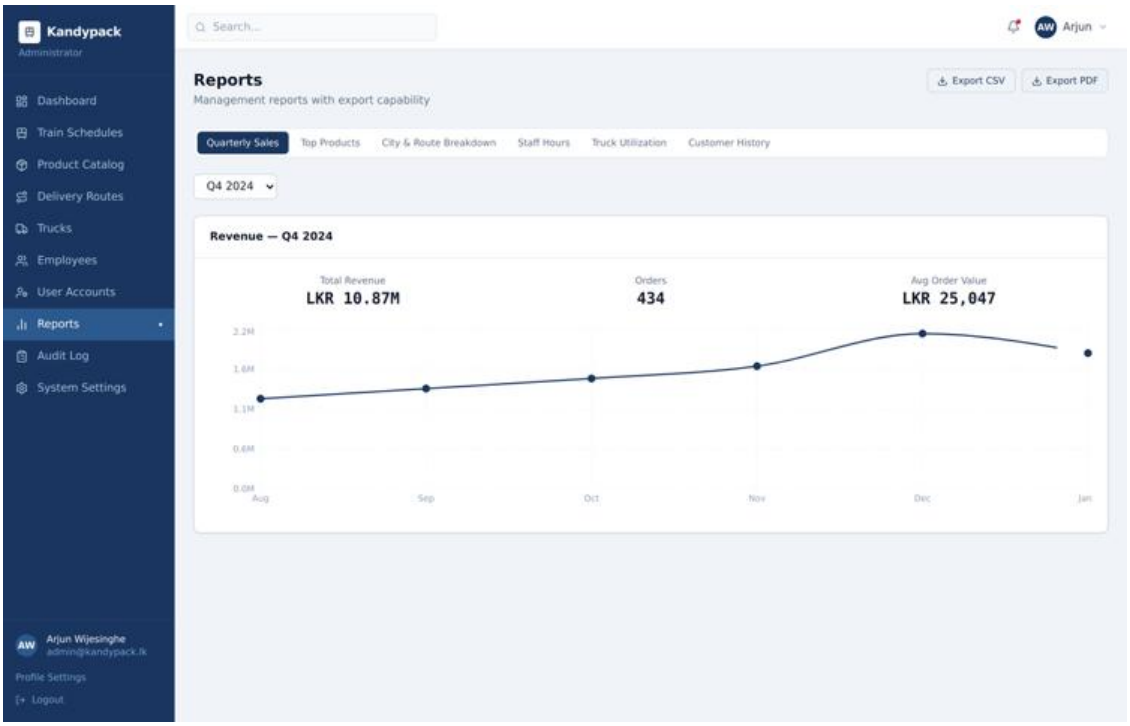


Figure 5: Report Overview screen, with access to all six management reports

3.1.3 Dispatcher Interfaces

- Train Loading screen: pending orders queued for a destination, a running total of how much of the upcoming trip's capacity has already been allocated to orders in the system (not a physical reading from the train itself), and a per-order loading-status control (Not Loaded / Partially Loaded / Loaded) so the Dispatcher can record physical loading as it happens.
- Weekly Roster screen: system-suggested truck/driver/assistant requirement per route for the coming week, with drag-and-drop or form-based assignment and inline conflict warnings (overlap, consecutive-run, weekly-hour breaches).

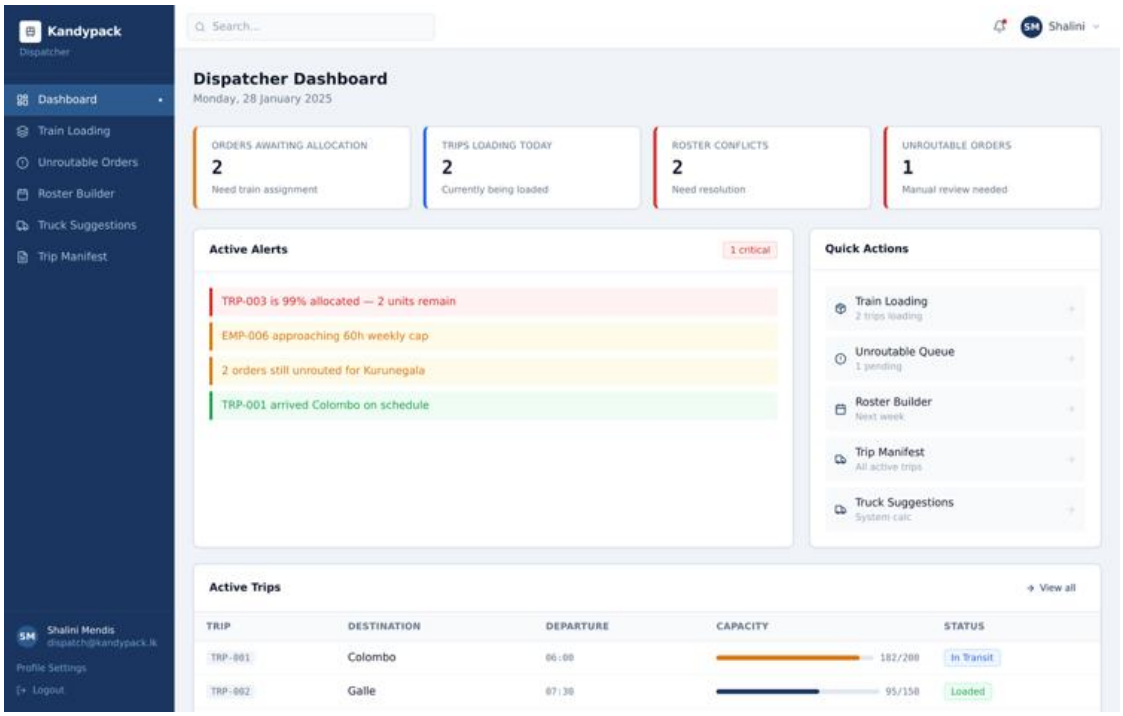


Figure 6: Dispatcher Dashboard (home screen), showing orders awaiting allocation, trips loading today, and roster conflicts

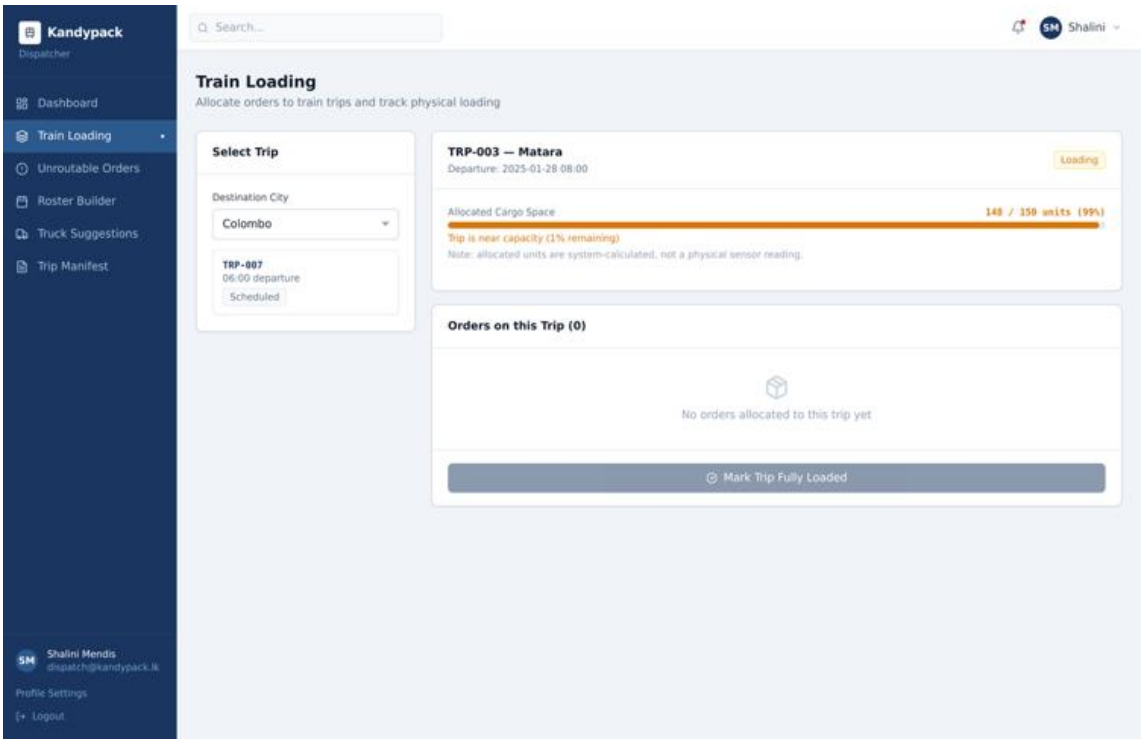


Figure 7: Train Loading screen, showing allocated capacity for the selected trip

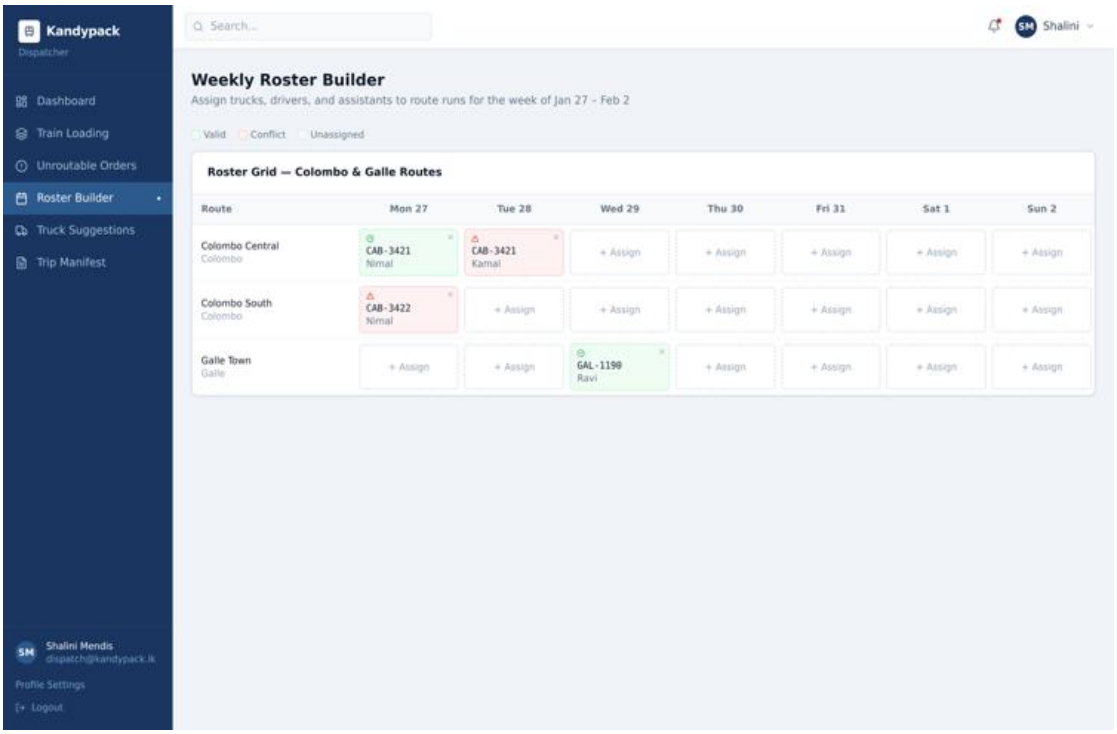


Figure 8: Weekly Roster Builder screen, with inline conflict status per assignment

3.1.4 Regional Store Manager Interfaces

- Incoming Shipment screen: expected vs. received quantity entry per train trip arrival, with a discrepancy flag and note field.
- Daily Dispatch screen: today's truck runs for routes covering the manager's city, and the orders loaded onto each run.

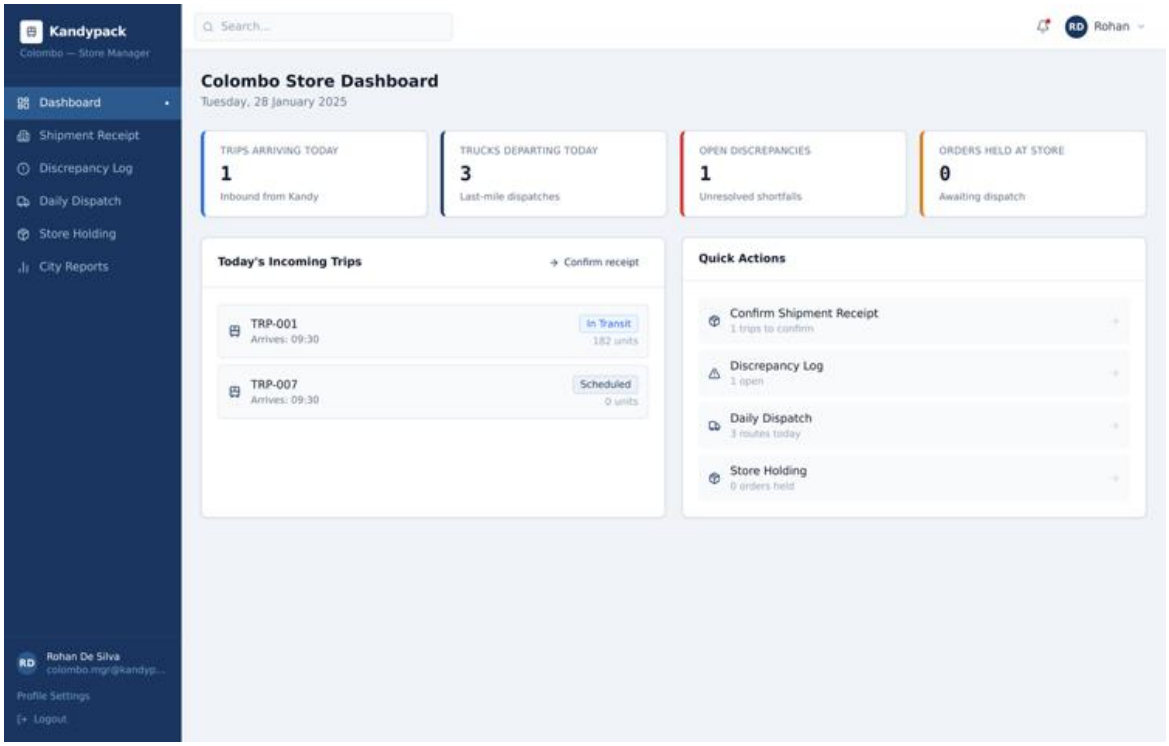


Figure 9: Regional Store Manager Dashboard (home screen), scoped to a single city (Colombo shown)

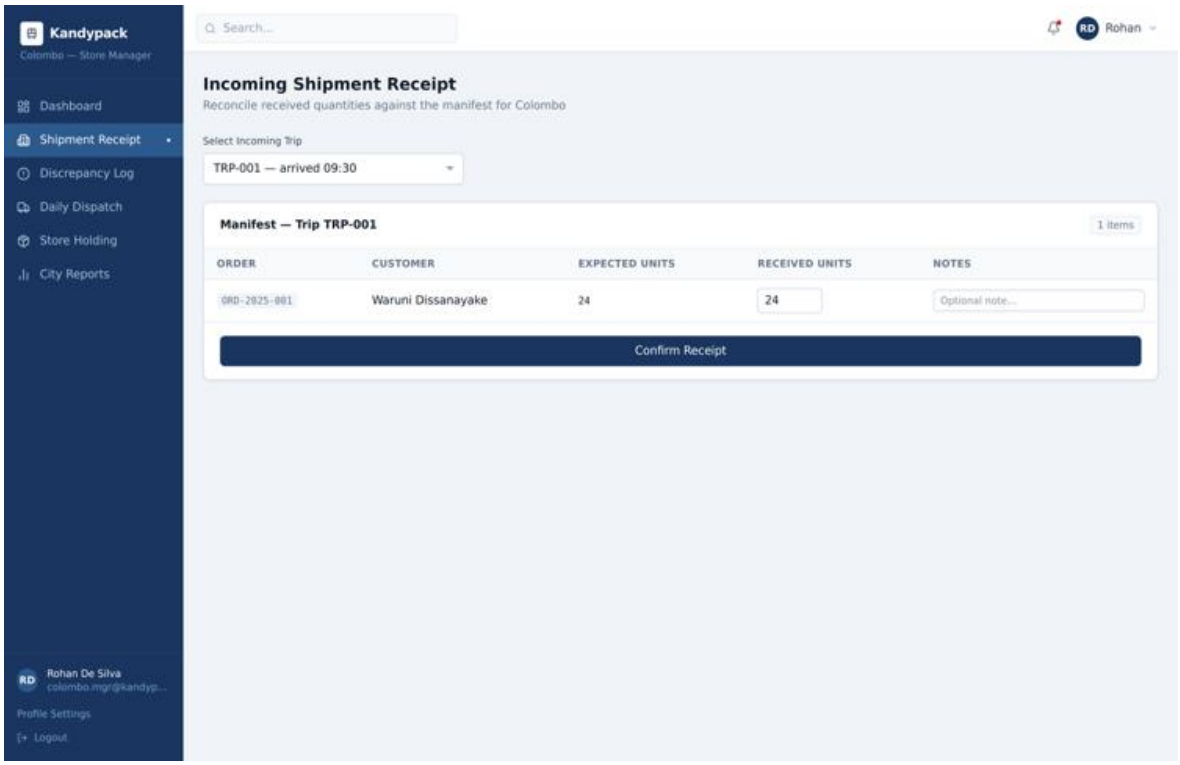


Figure 10: Incoming Shipment Receipt screen, for reconciling received quantities against the train manifest

3.1.5 Driver / Assistant Interface (simplified, mobile-first)

- Today's Run card: route name, truck ID, paired driver/assistant name, and ordered stop list.
- Single-tap actions: "Confirm Pickup", "Mark Delivered", "Report Issue" per stop, with minimal required text entry.

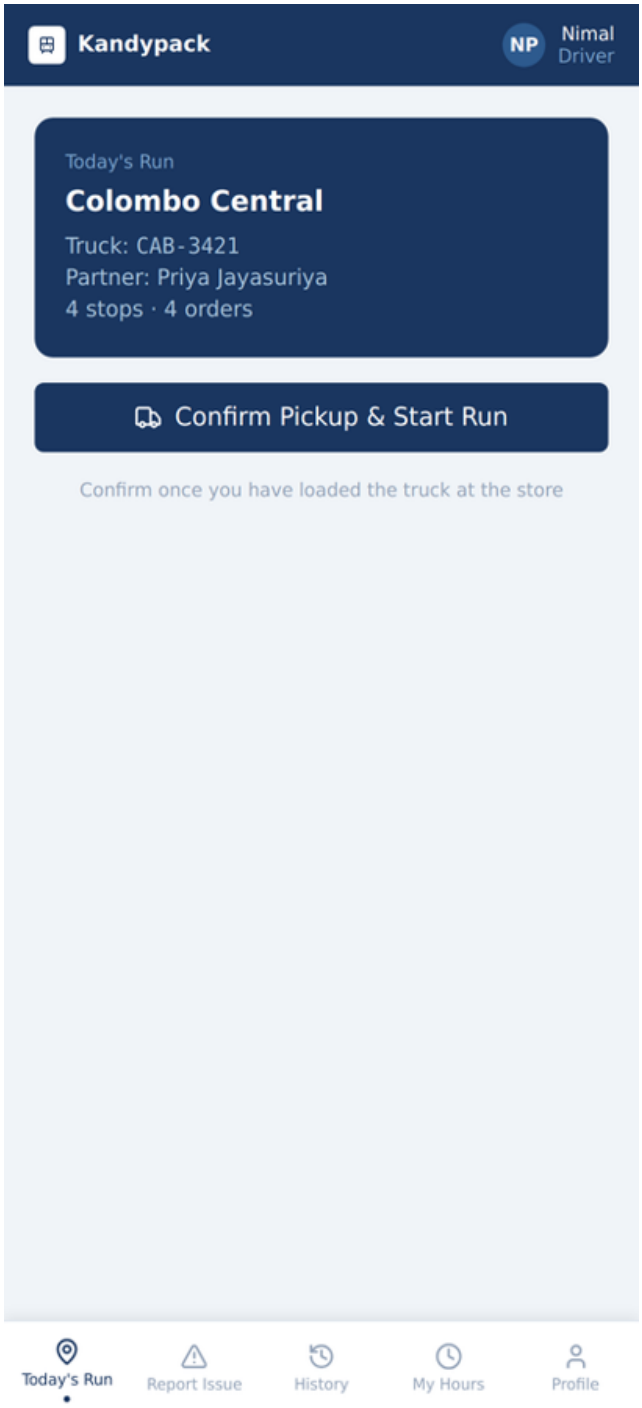


Figure 11: Today's Run screen (Mobile view), showing the assigned route, truck, partner, and stop list

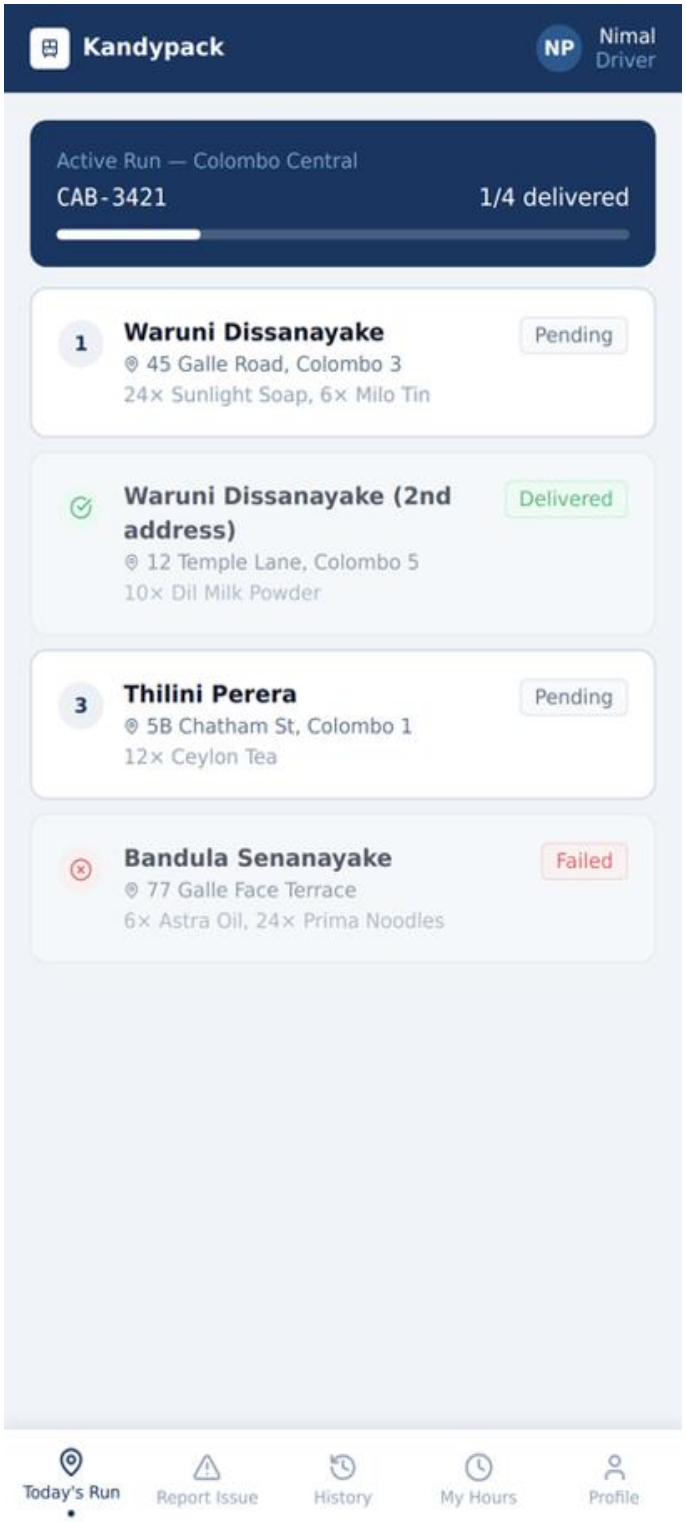


Figure 12: Stop Detail screen (Mobile view), showing a single stop's delivery actions

3.1.6 Customer Interfaces

- Order Placement form: item selection with quantity, delivery address, requested delivery date, with inline validation of the 7-day minimum lead time and route coverage.
- Order History list: status per order (Placed, Scheduled, In Transit, Delivered, Closed) with a "Confirm Receipt" action once a driver has marked delivery complete.

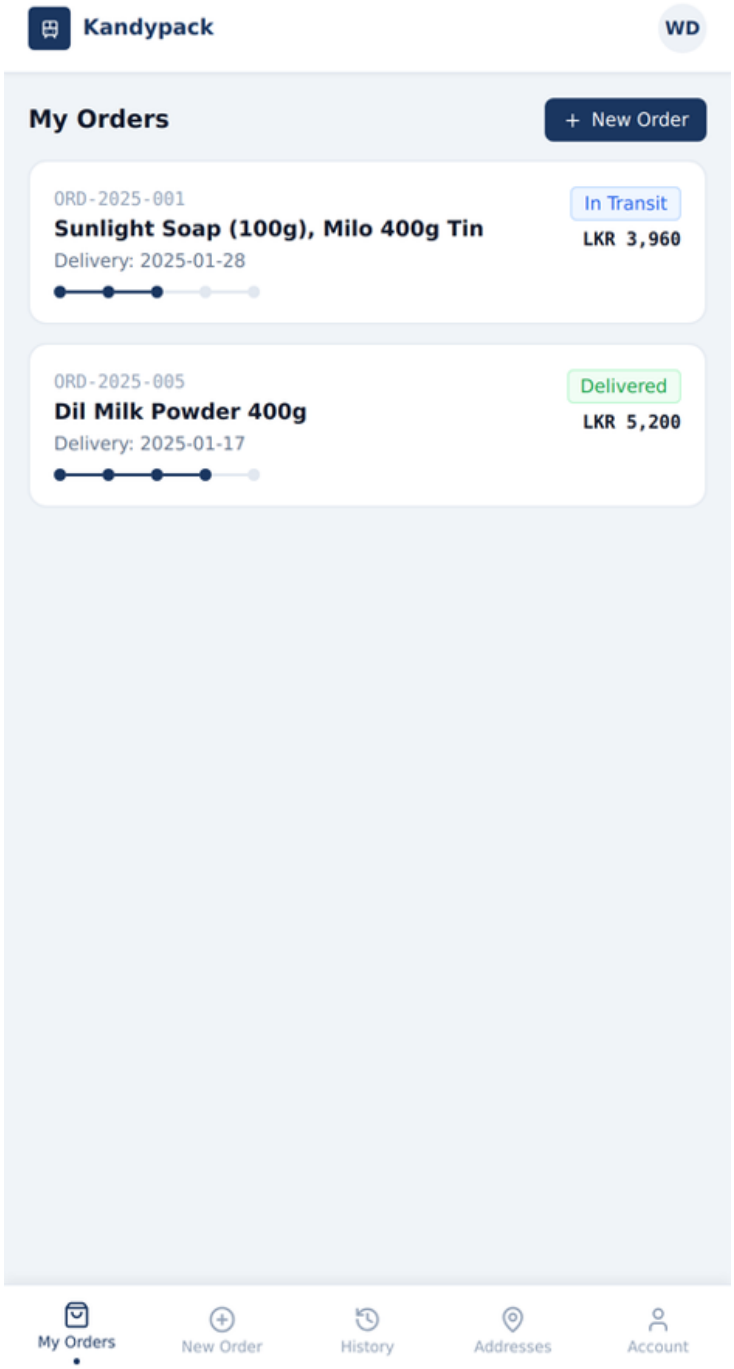


Figure 13: My Orders screen (mobile-friendly consumer layout), showing order status and tracking

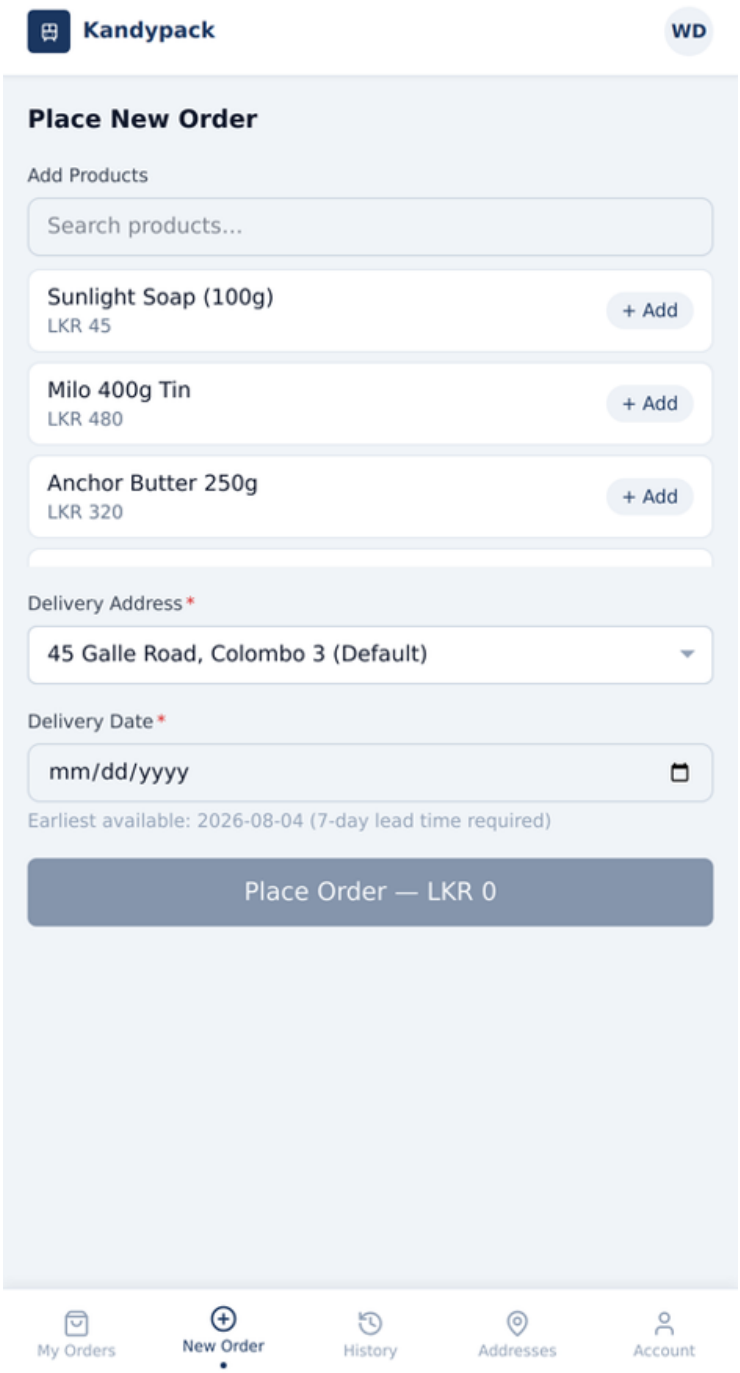


Figure 14: New Order placement screen (mobile-friendly consumer layout)

3.1.7 Common UI Standards

- Responsive layout adapting to desktop, tablet, and mobile screen sizes.
- Standard elements on every screen: search, account menu, help, and (for internal roles) a notifications/alerts panel.
- Errors surfaced via a non-blocking snack bar/toast at the bottom of the screen.
- Destructive or override actions (deleting a schedule, overriding a capacity/hour limit) require confirmation and, where noted in Section 5.2, a password re-prompt.

3.2 Hardware Interfaces

- Web Server: hosts the application logic and serves the API; must have sufficient memory/CPU to serve concurrent role-based sessions.
- Database Server: MySQL instance storing all order, schedule, roster, and reporting data.
- Client Devices: any desktop, tablet, or smartphone capable of running a supported browser (Section 3.3.2); no specialized hardware (e.g., barcode scanners, GPS trackers) is required for Version 1.0, though the schema should not preclude adding them later.

3.3 Software Interfaces

3.3.1 Supported Operating Systems

Windows 10 (21H2+), Windows 11 (22H2+), macOS Monterey/Ventura or later, and Linux (supported on Ubuntu 18.04+) for client browser access; the server runs on Linux.

3.3.2 Web Browser Compatibility

Google Chrome 60+, Microsoft Edge 80+, Mozilla Firefox 60+, Safari 12+.

3.3.3 Database Management System

MySQL, with support for foreign keys, transactions, stored procedures, functions, and triggers as required by Section 6.

3.3.4 Libraries and Frameworks

Front end: React with a responsive component library (e.g., Bootstrap or an equivalent). Back end: Node.js with Express for API and business-logic layers.

3.3.5 External Services

An SMS/email provider (e.g., Twilio) for order and roster notifications is assumed available; no external railway capacity API is used in Version 1.0 (see Section 2.5).

3.4 Communications Interfaces

- Protocol: HTTPS for all client-server communication (TLS 1.2 or higher).
- Architecture: RESTful API, JSON request/response bodies.
- Authentication: token-based (JWT or OAuth2) sessions.
- Notifications: outbound SMS/email via the external notification provider for delivery status changes, roster assignments, and capacity-delay alerts.

4. System Features

Functional requirements are organized by system feature, following the order goods move through the business: placement, rail transport, store receipt and truck planning, roster scheduling, last-mile execution, then the cross-cutting reporting, account, and notification features.

4.1 Customer Order Placement and Validation (Priority: High)

4.1.1 Description and Priority

Allows a customer to submit a multi-item order, enforces the 7-day advance-notice policy, and matches the delivery address to a predefined last-mile route. High priority: no downstream feature can function without a validated order.

4.1.2 Stimulus/Response Sequences

Stimulus: A customer submits a new order with item list, quantities, delivery address, and requested delivery date.

Response: the system computes total order value and volume, validates the 7-day lead time, matches the address to a route, and either creates a Placed order or rejects it with a specific reason (or flags it Unroutable for manual Administrator/Dispatcher review if no route covers the address).

4.1.3 Functional Requirements

REQ-ORD-01: The system shall accept an order containing one or more items (each with product reference and quantity), customer identity, requested delivery date, and delivery address.

REQ-ORD-02: The system shall reject any order whose requested delivery date is fewer than 7 days from the placement date, returning a descriptive validation error.

REQ-ORD-03: The system shall match the delivery address to a predefined delivery route; if no route covers the address, the order shall be flagged "Unroutable" and routed to Dispatcher/Administrator review rather than silently rejected.

REQ-ORD-04: The system shall compute and persist the total order value and total physical volume from item quantities, unit price, and each product's space-consumption rate.

REQ-ORD-05: The system shall maintain full order status history per customer: Placed -> Scheduled -> In Transit -> Delivered -> Closed, with timestamps for each transition.

REQ-ORD-06: The system shall allow a customer to modify or cancel an order only while it is in Placed status, i.e., before it has been allocated to a train trip; all changes shall be audit-logged.

REQ-ORD-07: Once a driver marks an order Delivered, the system shall require a separate customer confirmation ("Confirm Receipt") before the order transitions to Closed; if the customer has not confirmed within 48 hours of the Delivered timestamp, the system shall auto-close the order and log that it was system-closed.

4.2 Rail Cargo Scheduling and Train Loading (Priority: High)

4.2.1 Description and Priority

Schedules bulk transport of validated orders onto Sri Lanka Railways trips, respecting each trip's fixed cargo capacity and each product's space-consumption rate, splitting orders across trips where necessary, and recording the physical loading event performed by the Dispatcher at Kandy HQ.

4.2.2 Stimulus/Response Sequences

Stimulus: One or more validated orders are ready for bulk transit to a destination city.

Response: the system allocates order quantities to the earliest train trip(s) with sufficient remaining capacity, queuing any excess to the next available trip; the Dispatcher confirms physical loading, which locks the trip's manifest.

4.2.3 Functional Requirements

REQ-RAIL-01: The Administrator shall be able to define train trips between Kandy and each destination city (Colombo, Negombo, Galle, Matara, Jaffna, Trincomalee) with departure/arrival times and total cargo capacity.

REQ-RAIL-02: Each product shall have an Administrator-defined space-consumption rate used to compute the space an order quantity requires when allocating to a trip.

REQ-RAIL-03: The system shall assign order quantities to the earliest train trip(s) with sufficient remaining space for the destination city, splitting a single order across multiple trips when one trip's remaining capacity is insufficient.

REQ-RAIL-04: The system shall enforce trip capacity atomically within a database transaction; no trip may be overbooked beyond its defined capacity under concurrent allocation.

REQ-RAIL-05: Where an order (or part of an order) cannot be accommodated on its intended trip, the Dispatcher will schedule the overflow to the next available trip and record the resulting delay against the order.

REQ-RAIL-06: The Dispatcher shall be able to record, per order allocated to a trip, whether it is Not Loaded, Partially Loaded, or Loaded, reflecting the actual physical loading process at Kandy. The trip's manifest is locked and the trip's status becomes Loaded only once every order allocated to it has been marked Loaded.

REQ-RAIL-07: Where an order cannot be fully loaded by departure time despite having been allocated to that trip (e.g., a physical loading shortfall discovered at the platform), the system shall treat the unloaded portion the same way as capacity overflow (REQ-RAIL-05), manually queuing it to the next available trip and logging the change, rather than leaving it in an undefined state.

4.3 Store Receipt and Truck Requirement Planning (Priority: High)

4.3.1 Description and Priority

Handles arrival of goods at a destination store, reconciliation of received quantities against the train manifest, and system-suggested daily truck requirements per route based on historical order volume, to support the Dispatcher's weekly roster planning.

4.3.2 Stimulus/Response Sequences

Stimulus: A train trip marked Loaded arrives at its destination city.

Response: The Regional Store Manager records received quantities against the manifest; the system flags any shortfall/discrepancy and updates each order's status; the system also recalculates the suggested truck requirement for each route serving that store.

4.3.3 Functional Requirements

REQ-STORE-01: The system shall present the Regional Store Manager with the expected manifest for an arriving train trip and accept received-quantity entry per order line.

REQ-STORE-02: Where received quantity is less than manifest quantity, the system shall flag the shipment as "Discrepant", record the shortfall, and notify the Dispatcher/Administrator.

REQ-STORE-03: Upon full or partial receipt confirmation, the system shall update the status of each affected order accordingly. Items received through partial deliveries shall be stored until the remaining items are received, after which the complete order shall be made available for last-mile dispatch.

REQ-STORE-04: The system shall compute a suggested daily truck count per route as the rolling average number of orders for that route over a configurable trailing period, divided by average truck order-capacity, rounded up, with a minimum of 1 truck per route per operating day.

REQ-STORE-05: The Dispatcher shall be able to view and override the system-suggested truck count per route when building the weekly roster.

4.4 Truck, Driver, and Assistant Roster Scheduling (Priority: High)

4.4.1 Description and Priority

Assigns trucks with a driver and assistant to each route's daily run(s), one week ahead of the delivery date, while enforcing nonoverlap, consecutive-assignment, and weekly working-hour rules for drivers and assistants.

4.4.2 Stimulus/Response Sequences

Stimulus: The Dispatcher builds the roster for the coming week, using the suggested truck requirement per route from Feature 4.3. Response: the system validates each proposed assignment against roster rules in real time and either accepts it or returns a specific conflict/violation for resolution.

4.4.3 Functional Requirements

REQ-SCHED-01: The system shall ensure no truck, driver, or assistant is assigned to overlapping routes or time windows.

REQ-SCHED-02: The system shall prevent a driver from being scheduled for two consecutive truck deliveries without an intervening rest break of at least 2 hours; this minimum duration shall be stored as a configurable system parameter rather than hard-coded, so it can be revised by the Administrator without a code change.

REQ-SCHED-03: The system shall allow an assistant to be scheduled for at most two consecutive routes; any proposed third consecutive assignment shall be blocked or flagged for override.

REQ-SCHED-04: The system shall enforce weekly working-hour limits of 40 hours for drivers and 60 hours for assistants, preventing further scheduling beyond these limits unless explicitly overridden with a logged justification.

REQ-SCHED-05: The system shall allow the Dispatcher to override a scheduling rule with a required justification, and shall log all overrides for audit.

REQ-SCHED-06: The roster shall be created at least one week (7 days) ahead of the delivery date it covers.

4.5 Last-Mile Delivery Execution and Status Tracking (Priority: High)

4.5.1 Description and Priority

Covers the Driver/Assistant's day-of-delivery workflow: pickup confirmation at the store, per-stop delivery outcome recording, and issue/delay reporting, plus the customer-side receipt confirmation from Feature 4.1.

4.5.2 Stimulus/Response Sequences

Stimulus: A driver/assistant begins their assigned run.

Response: the system records pickup confirmation, then accepts a delivered/failed outcome per stop, updating order status and triggering the customer confirmation step for successful deliveries.

4.5.3 Functional Requirements

REQ-DEL-01: The system shall present the Driver and Assistant with their assigned run for the day: route, truck, paired staff member, and ordered stop list.

REQ-DEL-02: The system shall record a pickup confirmation from the store before any stop on that run can be marked delivered.

REQ-DEL-03: The system shall allow the Driver or Assistant to record each stop's outcome as Delivered, Failed, or Attempted, with an optional note.

REQ-DEL-04: Where a stop is marked Failed, the system shall automatically re-queue the order for the next available run on the same route rather than leaving it unassigned.

REQ-DEL-05: The system shall track individual order item delivery status (Pending, Out for Delivery, Delivered) visible to the customer and the Regional Store Manager.

4.6 Reporting Module (Priority: Medium-High)

4.6.1 Description and Priority

Generates the six management reports required by Kandypack for performance, usage, and compliance monitoring, each filterable and scoped by user role.

4.6.2 Stimulus/Response Sequences

Stimulus: An authorized user requests a report with optional filters (quarter, city, route, driver).

Response: the system retrieves, aggregates, and presents the requested data, restricted to the requester's data scope (e.g., a Regional Store Manager sees only their city).

4.6.3 Functional Requirements

REQ-RPT-01: The system shall generate a quarterly sales report showing total value and total volume of goods sold.

REQ-RPT-02: The system shall list the most-ordered items for a specified quarter, ranked by quantity and by value.

REQ-RPT-03: The system shall provide city-wise and route-wise sales breakdowns for a given period.

REQ-RPT-04: The system shall report driver and assistant working hours per week/month and flag any violations of the limits defined in REQ-SCHED-04.

REQ-RPT-05: The system shall generate a monthly truck utilization report (trips per truck, idle time, capacity usage).

REQ-RPT-06: The system shall display an individual customer's full order history with delivery timeline and status updates.

REQ-RPT-07: All reports shall be exportable to CSV and PDF and restricted to the requesting user's data scope (Administrator: all data; Regional Store Manager: own city only; Customer: own orders only).

4.7 User Management and Authentication (Priority: Medium)

4.7.1 Description and Priority

Controls system access and secures operations across the six user classes defined in Section 2.3.

4.7.2 Stimulus/Response Sequences

Stimulus: A user attempts to log in or perform a protected action.

Response: the system authenticates identity, authorizes the action against the user's role and data scope, and logs the access/action.

4.7.3 Functional Requirements

REQ-USR-01: The system shall support role-based access control with at least six roles: Administrator, Dispatcher, Regional Store Manager, Driver, Assistant, and Customer.

REQ-USR-02: The system shall require authentication before any operation other than customer self-registration and public login.

REQ-USR-03: Only the Administrator shall be able to create or deactivate Dispatcher, Regional Store Manager, Driver, and Assistant accounts; Customers shall self-register.

REQ-USR-04: A Regional Store Manager's data access shall be restricted to their assigned city; a Customer's data access shall be restricted to their own orders.

REQ-USR-05: The system shall record an audit log of user actions affecting orders, schedules, and rule overrides, including who made the change and when.

4.8 Notifications and Alerts (Priority: Medium)

4.8.1 Description and Priority

Keeps stakeholders informed of scheduling conflicts, capacity delays, discrepancies, and delivery outcomes.

4.8.2 Stimulus/Response Sequences

Stimulus: A constraint violation or significant state change occurs (e.g., train overcapacity delay, discrepant receipt, driver approaching a weekly-hour limit).

Response: the system generates and delivers an alert to the relevant role via the in-app notifications panel and, where configured, SMS/email.

4.8.3 Functional Requirements

REQ-NOT-01: The system shall alert the Regional Store Manager and Dispatcher when a shipment is received with a discrepancy (REQ-STORE-02).

REQ-NOT-02: The system shall flag an approaching driver/assistant weekly-hour limit before it is exceeded.

REQ-NOT-03: The system shall notify the Dispatcher of any unresolved scheduling conflict or unassigned route run.

REQ-NOT-04: The system shall notify a customer by SMS/email when their order's status changes to Scheduled, In Transit, or Delivered.

REQ-NOT-05: The system shall allow users to acknowledge or dismiss alerts, with the acknowledgment logged.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- The system shall support multiple concurrent users across all roles without noticeable performance degradation.
- Standard UI interactions (e.g., button clicks, form submission) shall respond within 1 second under normal load.
- Standard management reports shall generate within 10 seconds for typical (single-quarter, single-city) data volumes.
- Order validation and initial train-trip allocation shall complete within 3 seconds for orders of fewer than 10 line items.
- The system shall support at least 50 simultaneous order submissions without failure or unacceptable lag, to allow for future growth.

5.2 Safety Requirements

- The system shall perform automatic backups twice daily (midday and end of operating day) to minimize data loss.
- In the event of a crash or power failure, the system shall be able to restore from the most recent backup without loss of data integrity.
- Deleting or editing a published train, roster, or route schedule shall require password re-authentication and display a confirmation warning; the same applies to any action that overrides a capacity or working-hour limit.
- The system shall detect and block overlapping truck/driver/assistant assignments before they can be saved (see REQ-SCHED-01).

5.3 Security Requirements

- Access shall require authentication with a password meeting a minimum length and complexity policy.
- All operations shall be authorized against the user's role and data scope (role-based access control).
- Passwords shall be stored hashed; all data in transit shall use HTTPS/TLS.
- The system shall be designed to resist common web threats, including SQL injection and cross-site scripting (XSS).
- All user actions affecting orders, schedules, and overrides shall be logged for audit and traceability (see REQ-USR-05).

5.4 Software Quality Attributes

- Availability: the system shall maintain 99% uptime during operational hours (6 AM - 10 PM); updates shall be applied outside these hours.
- Usability: interfaces shall require minimal training, particularly the Driver/Assistant simplified view, and shall run acceptably on modest hardware.
- Maintainability: the codebase shall be modular and documented to support future extension (e.g., warehouse integration, payment processing).
- Scalability: the architecture shall accommodate additional products, routes, and users without a full redesign.
- Reliability: the system shall perform consistently under defined load, with a target failure rate below 0.1%.
- Portability: the application shall run on all browsers listed in Section 3.3.2 and be compatible with standard relational database systems.

5.5 Business Rules

- Orders must be placed at least 7 days before the requested delivery date (REQ-ORD-02).
- A driver may not be scheduled for two consecutive truck deliveries without an intervening rest break of at least 2 hours; weekly working time must not exceed 40 hours (REQ-SCHED-02, REQ-SCHED-04).
- An assistant may be scheduled for at most two consecutive deliveries, with a weekly limit of 60 hours (REQ-SCHED-03, REQ-SCHED-04).
- No truck, driver, or assistant may be assigned to overlapping delivery routes (REQ-SCHED-01).
- If an order exceeds a train trip's cargo capacity, the excess must be automatically rescheduled to the next available trip, respecting space constraints (REQ-RAIL-05).
- Every delivery must be matched to the correct predefined route based on the delivery address (REQ-ORD-03).
- A delivery is only considered Closed once both the driver has marked it Delivered and the customer has confirmed receipt, or the 48-hour auto-close window has elapsed (REQ-ORD-07).
- A Regional Store Manager may only view and act on data for their own assigned city (REQ-USR-04).

6. Other Requirements

Database Requirements: the system shall use a relational database (MySQL) with full support for ACID transactions, primary and foreign keys, indexing, stored procedures, functions, and triggers. At minimum, the following areas require procedural logic to guarantee consistency:

- A stored procedure/trigger enforcing atomic train-trip capacity checks at allocation time (REQ-RAIL-03, REQ-RAIL-04), to prevent race conditions when multiple orders are allocated concurrently.
- A trigger cascading order status transitions (Placed -> Scheduled -> In Transit -> Delivered -> Closed) so downstream tables (roster, reporting aggregates) stay consistent.
- A stored procedure enforcing roster rules (nonoverlap, consecutive-assignment, weekly-hour caps) atomically at assignment time (REQ-SCHED-01 to REQ-SCHED-04), rather than relying solely on application-layer checks.
- A trigger or scheduled procedure recomputing the suggested truck requirement per route (REQ-STORE-04) as new order data arrives.
- Indexes on foreign keys and on frequently filtered/reported columns: order delivery date, destination city, route ID, and driver/assistant ID, to keep the Section 5.1 reporting-time targets achievable.

Test Data: the database shall be populated with at least 40 orders, 10 distinct delivery routes, and a valid, capacity-consistent train schedule sufficient to exercise the allocation, splitting, rostering, and reporting logic described in Section 4.

Internationalization: Version 1.0 is English-only; the schema and UI text shall be structured so that multilingual support can be added later without redesign.

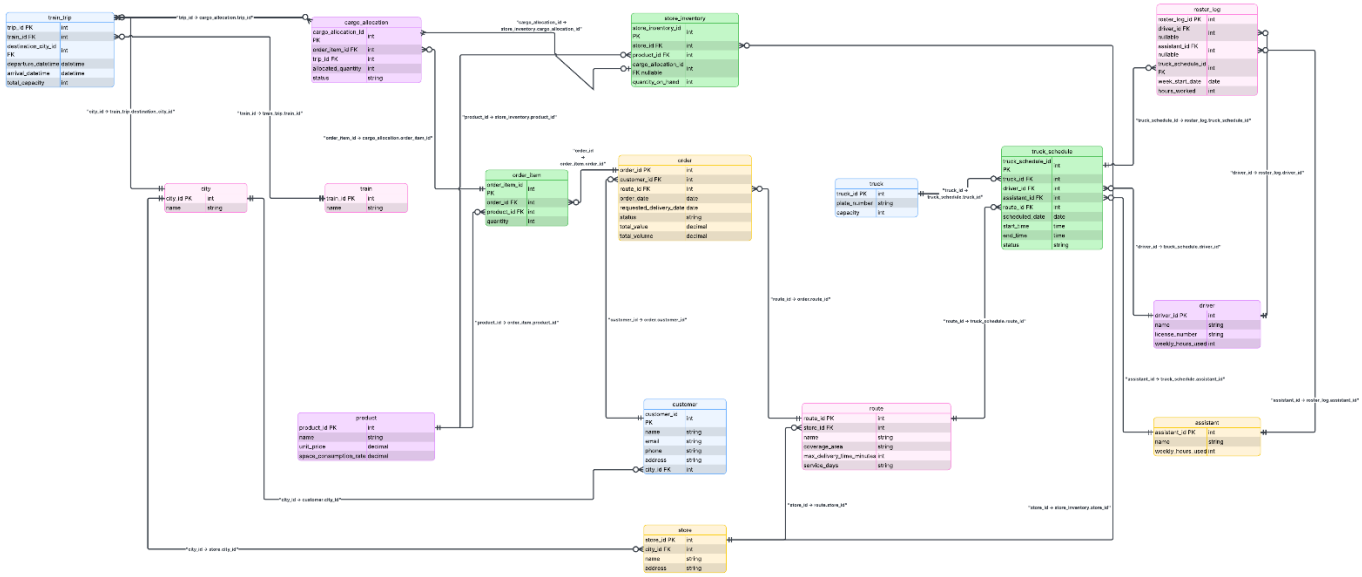
Legal Requirements: the system shall comply with Sri Lankan data protection expectations regarding storage and access to customer personal data.

Reuse Objectives: the codebase shall be modular enough to support future integration with external shipping systems or a dedicated mobile application, without requiring a rewrite of the core order/rail/roster logic.

Appendix A: Glossary

Term	Definition
FMCG	Fast-Moving Consumer Goods
RBAC	Role-Based Access Control
Trip	A single scheduled train journey between Kandy and a destination city, with a fixed cargo capacity
Run	A single truck's scheduled delivery trip along one route on a given day
Route	A predefined coverage area served by last-mile trucks, with a maximum delivery time
Manifest	The list of orders/quantities allocated to a specific train trip
Unroutable	An order whose delivery address is not covered by any predefined route
Discrepant	A shipment where received quantity does not match the manifest quantity
ACID	Atomicity, Consistency, Isolation, Durability - the transactional guarantees required of the database

Appendix B: Entity-Relationship Diagram



Appendix C: To Be Determined List

- TBD-1: Full permission matrix mapping each of the six roles to specific screen/action-level permissions.
- TBD-2: Error handling and logging mechanism details (log retention period, log storage location).
- TBD-3: Implementation timeline and milestone schedule.
- TBD-4: Whether the Driver/Assistant view should be built as an installable Progressive Web App for offline tolerance, or remain a standard responsive page.

Appendix D: Team Members

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